Software Requirements Specification (SRS)

Library Automation System

1. Introduction

1.1 Purpose

This document specifies the requirements of the **Library Automation System** to be deployed in an academic library. The system will be used by librarians, students, and faculty members to manage book search, issue, return, and renewal operations.

The intended audience of this document includes system analysts, developers, testers, and library administrators.

1.2 Project Scope

The scope of the Library Automation System is to automate the **core library operations**, including:

- Book catalog management
- Book issue, return, and renewal
- User account management
- Availability queries

Activities such as advanced analytics, e-book integration, and inter-library loan services are **outside the current scope** and may be considered during future evolution of the system.

1.3 Environmental Characteristics

The system will operate in the following environment:

- Hardware: Desktop computers and servers within the library
- Software: Web browser, database management system
- Network: Local Area Network (LAN) and Internet for web-based access

2. Overall Description of the SRS Document

2.1 Product Perspective

The Library Automation System is a **replacement for the existing manual system**. It operates as a standalone system but may later be integrated with institutional systems such as student information systems.

At a high level, the system consists of:

- User Interface Module
- Library Management Module
- Database Module

2.2 Product Features

The major features of the system include:

- Searching books by title, author, or ISBN
- Issuing and returning books
- Renewing borrowed books
- Viewing book availability

Detailed descriptions of these features are provided in **Section 3 (Functional Requirements)**.

2.3 User Classes

User Class	Description
Librarian	Manages books, users, and transactions
Student	Searches books, borrows and renews books
Faculty	Similar privileges as students, with extended borrowing period

2.4 Operating Environment

- Operating System: Windows or Linux
- Database: Relational DBMS
- Browser: Any modern web browser
- Server: Web application server

2.5 Design and Implementation Constraints

- The system shall use a **free or open-source DBMS**
- The system shall comply with institutional data privacy policies
- The system shall follow standard coding and documentation conventions
- Web-based architecture shall be used

2.6 User Documentation

The following documents will be provided:

- User manual for librarians
- User guide for students and faculty ,Online help and FAQs

3. Functional Requirements

R.1 Search Book

(Invoked by: Student, Librarian)

Description:
Once the user selects the search option, the system prompts the user to enter search keywords. The system searches the book database based on the entered keywords and displays the details of all books whose title or author name matches any of the keywords entered.The displayed details include title, author name, publisher name, year of publication, ISBN number, catalog number, number of copies available, and the location in the library.

R.1.1 Select search option

- **Input:** “Search Book” option
- **Output:** Prompt to enter search keywords

R.1.2 Search and display

- **Input:** Keywords
- **Output:** Matching book details
- **Processing:**
  Search the book list using keyword matching on title and author fields.

R.2 Issue Book

(Invoked by: Librarian)

Description:
Allows the librarian to issue a book to a registered student. The system verifies the eligibility of the student and the availability of the book. If the issue is successful, the system records the transaction and prints an issue receipt.

R.2.1 Select issue book option

- **Input:** “Issue Book” option
- **Output:** Prompt to enter student ID and book ID

R.2.2 Enter student details

- **Input:** Student ID
- **Output:** List of books currently issued to the student

- **Processing:**
  Retrieve borrowing records of the student.

  Note :Processing is written only when the system performs non-obvious logic; for simple input–output actions, it is safely omitted.

R.2.3 Enter book details

- **Input:** Book ID
- **Output:** Availability status of the book

R.2.4 Validate issue conditions

- **Processing:**
  Check whether the student has reached the maximum borrowing limit and whether the book is available.

R.2.5 Confirm issue

- **Output:** Issue confirmation message with due date
- **Processing:**
  Update borrowing records and assign the due date.

R.2.6 Print issue receipt

- **Output:** Issue receipt
- **Processing:**
  Generate and print the issue receipt containing student ID, book ID, issue date, and due date.

R.3 Return Book

(Invoked by: Student, Librarian)

Description:
Records the return of a borrowed book. If the book is returned after the due date, the system calculates the applicable fine.

R.3.1 Select return book option

- **Input:** “Return Book” option
- **Output:** Prompt to enter book ID

R.3.2 Enter book details

- **Input:** Book ID
- **Output:** Borrower details and due date displayed

R.3.3 Validate return date

- **Processing:**
  Compare the return date with the due date.

  Note :No explicit input/output is written because this step is a pure internal processing step. The required data is already available from previous interactions, and the result is used only to decide the next scenario.

R.3.4 return (no fine case)

- **Condition:** Book returned on or before due date
- **Processing:**
  Update borrowing records and mark the book as available.
- **Output:** Return confirmation message

R.3.5 Invoke fine calculation (fine case)

- **Condition:** Book returned after due date
- **Next function:** R.8 Calculate Fine

R.4 Renew Book

(Invoked by: Student)

Description:
Allows a student to renew a borrowed book, provided the book has not been reserved by another student.

R.4.1 Select renew option

- **State:** Student logged in
- **Input:** “Renew Book” option
- **Output:** Prompt for credentials

R.4.2 Authenticate student

- **Input:** Student ID and password
- **Output:** List of borrowed books

- **Processing:**
  Validate credentials and retrieve borrowing records.

R.4.3 Renew selected book

- **Input:** Selected book ID
- **Output:** Renewal confirmation or rejection message
- **Processing:**
  Check reservation status and update due date.

R.5 Check Due Books

(Invoked by: Student, Librarian)

Description:
Displays books that are due or overdue.

R.5.1 Select due book option

- **Input:** “Check Due Books” option
- **Output:** List of due or overdue books

R.5.2 Display due details

- **Output:** Book title, borrower name, due date, overdue days
- **Processing:**
  Compare due dates with the current date.

R.6 Manage Book Records

(Invoked by: Librarian)

Description:
Allows the librarian to maintain library book inventory records.

R.6.1 Add new book

- **Input:** Book details
- **Output:** Book registration confirmation
- **Processing:**
  Generate a unique book ID and store book details.

R.6.2 Update book details

- **Input:** Modified book information
- **Output:** Update confirmation

R.6.3 Delete book record

- **Input:** Book ID
- **Output:** Deletion confirmation

R.7 Manage Member Records

(Invoked by: Librarian)

Description:
Allows the librarian to manage student membership records.

R.7.1 Register member

- **Input:** Student details
- **Output:** Registration confirmation

R.7.2 Update member details

- **Input:** Updated information
- **Output:** Update confirmation

R.7.3 Deactivate member

- **Input:** Student ID
- **Output:** Deactivation confirmation

R.8 Calculate Fine

(Invoked by: System / Accountant)

Description:
Calculates fine for overdue books based on the number of overdue days and the applicable fine rate.

R.8.1 Retrieve overdue details

- **Input:** Book ID / Student ID
- **Processing:**
  Determine overdue days using return date and due date.

R.8.2 Compute fine amount

- **Output:** Fine amount
- **Processing:**
  Fine = Overdue days × Fine per day rate.

R.8.3 Print fine receipt

- **Output:** Fine receipt
- **Processing:**
  Generate and print fine receipt with overdue and payment details.

R.9 Check Book Availability*(Invoked by: Student, Librarian)*

Description:
Displays the availability status of a specific book.

R.9.1 Enter book details

- **Input:** Book ID / Book title
- **Output:** Availability status and number of copies

4. Non-Functional Requirements

N.1: Database
A database management system that is available **free of cost in the public domain** shall be used to store book records, member details, issue/return transactions, and fine information.

N.2: Platform
The software shall be developed to run on **both Windows and UNIX/Linux operating systems**.

N.3: Web Support
It shall be possible for students to **search book availability through a web browser** from any location within the institute network.

N.4: Performance
The system shall respond to book search and availability queries within **2 seconds** under normal operating conditions.

N.5: Security
Access to librarian and accountant functionalities shall be protected using **user authentication mechanisms** (login ID and password).

N.6: Reliability
The system shall ensure that **issue, return, and fine calculation records are not lost** in case of system failure.